other UML tool you are familiar with. The contract diagram is a convenient artifact for design discussions with stakeholders and the development team.

2.3 Development of a smart contract code

You are now ready to develop the smart contract code in a high-level language. Although many languages are available--such as Java, Python, and Go--these are general-purpose languages with rich syntax and semantics, supported by extensive libraries. For smart contract development, you need a limited language customized for blockchain operations. Solidity is one such language. The Ethereum foundation introduced this language, but other blockchain platforms, such as Hyperledger, also support it. You will use Solidity to code your smart contracts.

It’s important to understand that blockchain programming is not about porting or translating code written in a traditional high-level language to Solidity. Writing a contract code is about precise instructions for blockchain-oriented recording. Smart contract code does not need all the bells and whistles of a general-purpose language. On the other hand, it requires specific blockchain-oriented features for handling things like account addresses, rule specifications, and transaction reversals to be built into the language. Moreover, the smart contract code executes in a restricted sandbox environment to maintain consistency among the nodes of the blockchain. These are the justifications for using a